

Technical Project Report
On
“Customer Churn Business Analysis”

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To
The Developer Arena

1. Introduction

Customer churn is one of the biggest challenges for any business. Customer churn means that existing customers stop using a company's products or services. Losing customers reduces company revenue, increases marketing costs, and negatively affects business growth.

The purpose of this project is to perform a complete business analysis using customer churn data. The analysis helps identify important customer behavior patterns and provides data-driven recommendations that can help businesses improve customer retention and customer satisfaction.

This project follows the complete Data Analysis Life Cycle including data collection, cleaning, exploratory analysis, visualization, statistical analysis, and business recommendations.

2. Business Problem

Many organizations lose valuable customers every month due to poor customer experience, high monthly charges, lack of engagement, or better offers from competitors.

The company wants answers to the following questions:

- Why are customers leaving?
- Which customers are more likely to churn?
- Which factors influence customer churn?
- How can customer retention be improved?

This analysis helps management make better business decisions using data.

3. Project Objectives

The main objectives of this project are:

- Analyze customer churn data.
- Clean and preprocess the dataset.
- Perform Exploratory Data Analysis (EDA).
- Create professional visualizations.
- Perform correlation analysis.
- Conduct hypothesis testing.
- Build a regression model.
- Generate business insights.
- Provide actionable recommendations.

4. Dataset Description

Dataset Name:

Customer Churn Dataset

Dataset Size:

- Rows: 500
- Columns: 4

Main Features:

- Customer ID
- Monthly Charges
- Tenure
- Churn Status

The dataset contains customer information that helps understand customer behavior and identify factors affecting churn.

5. Tools and Technologies

Programming Language

- Python

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy
- Scikit-Learn

Development Environment

- Jupyter Notebook
- Visual Studio Code
- GitHub

6. Methodology

The project followed these steps:

Step 1

Business Problem Identification

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Step 2

Dataset Collection



Step 3

Data Cleaning



Step 4

Exploratory Data Analysis



Step 5

Visualization



Step 6

Statistical Analysis



Step 7

Business Insights



Step 8

Recommendations



Step 9

Final Documentation

7. Data Cleaning

The following data cleaning activities were performed:

- Checked missing values.
- Removed duplicate records.
- Verified data types.
- Prepared cleaned dataset.
- Saved cleaned_data.csv.

Data cleaning improves data quality and ensures accurate analysis.

8. Exploratory Data Analysis

EDA was performed to understand the dataset.

The following analyses were completed:

- Dataset Shape
- Dataset Information
- Descriptive Statistics
- Missing Values
- Duplicate Records
- Customer Churn Distribution

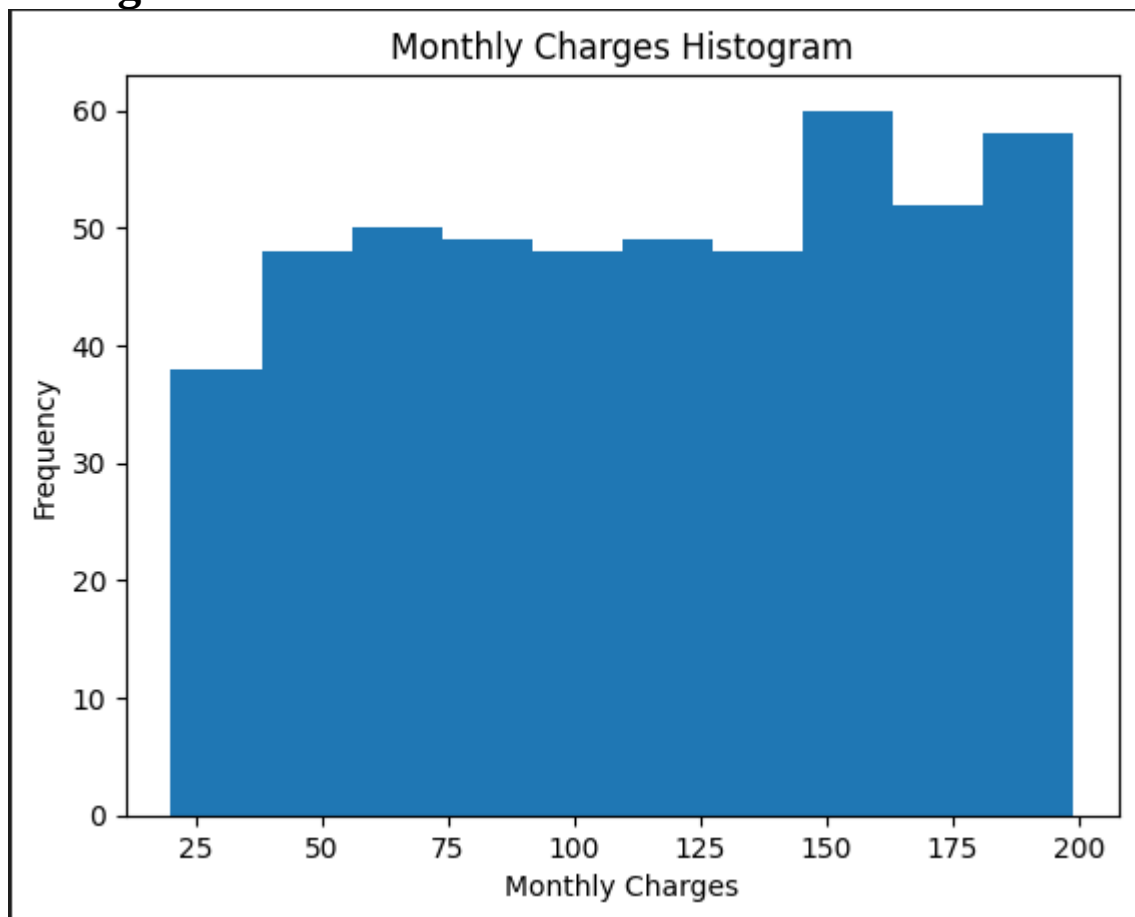
EDA helps identify trends and understand customer behavior before advanced analysis.

9. Data Visualization

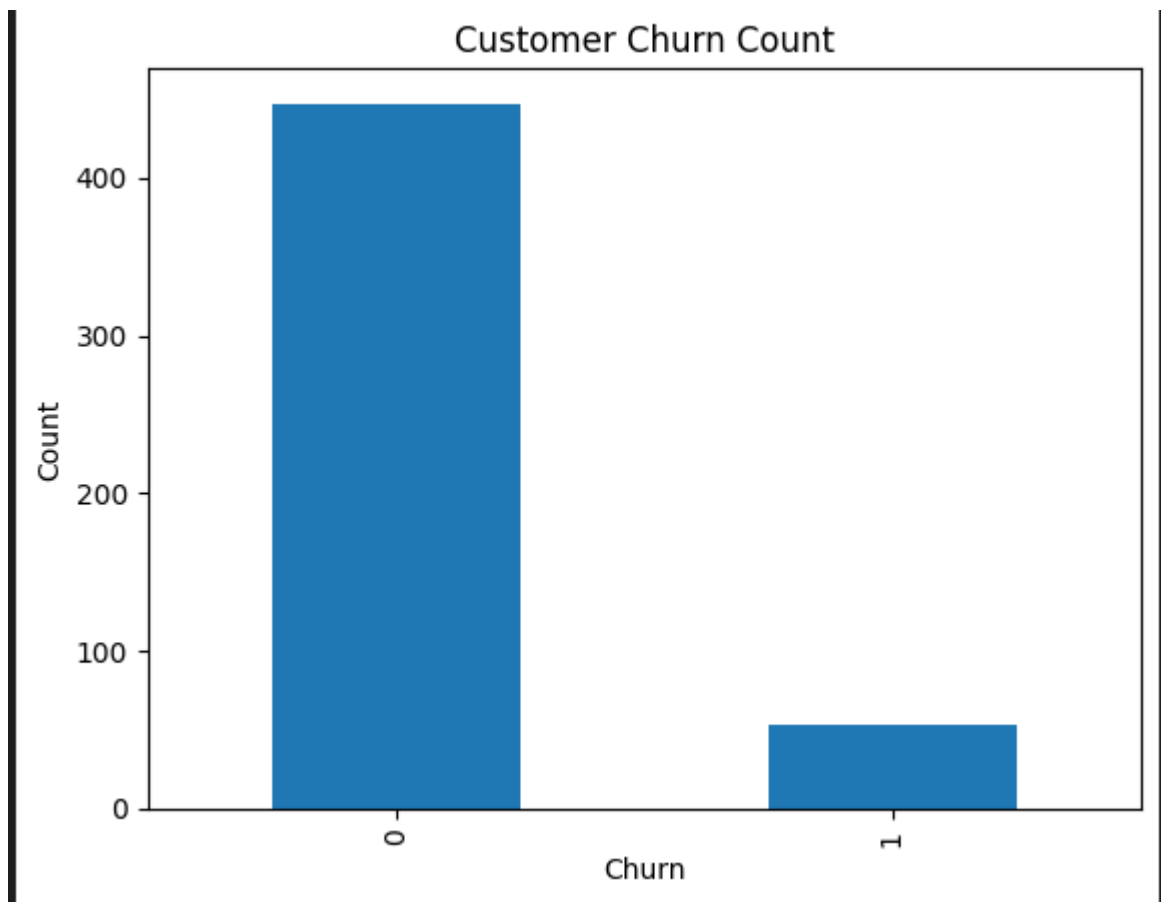
Several charts were created to understand customer behavior.

Visualizations included:

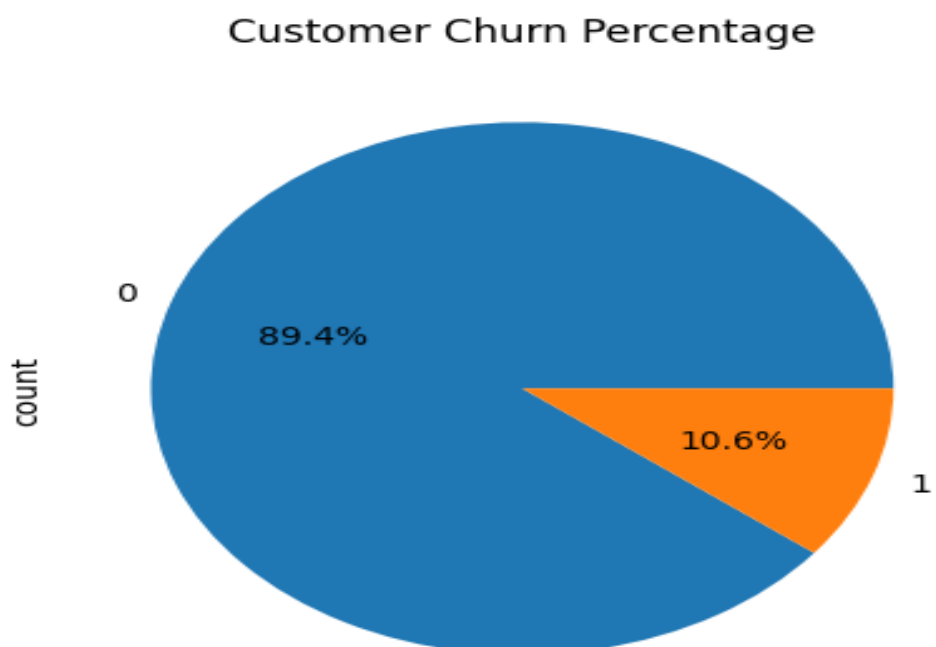
- **Histogram**



- Bar Chart

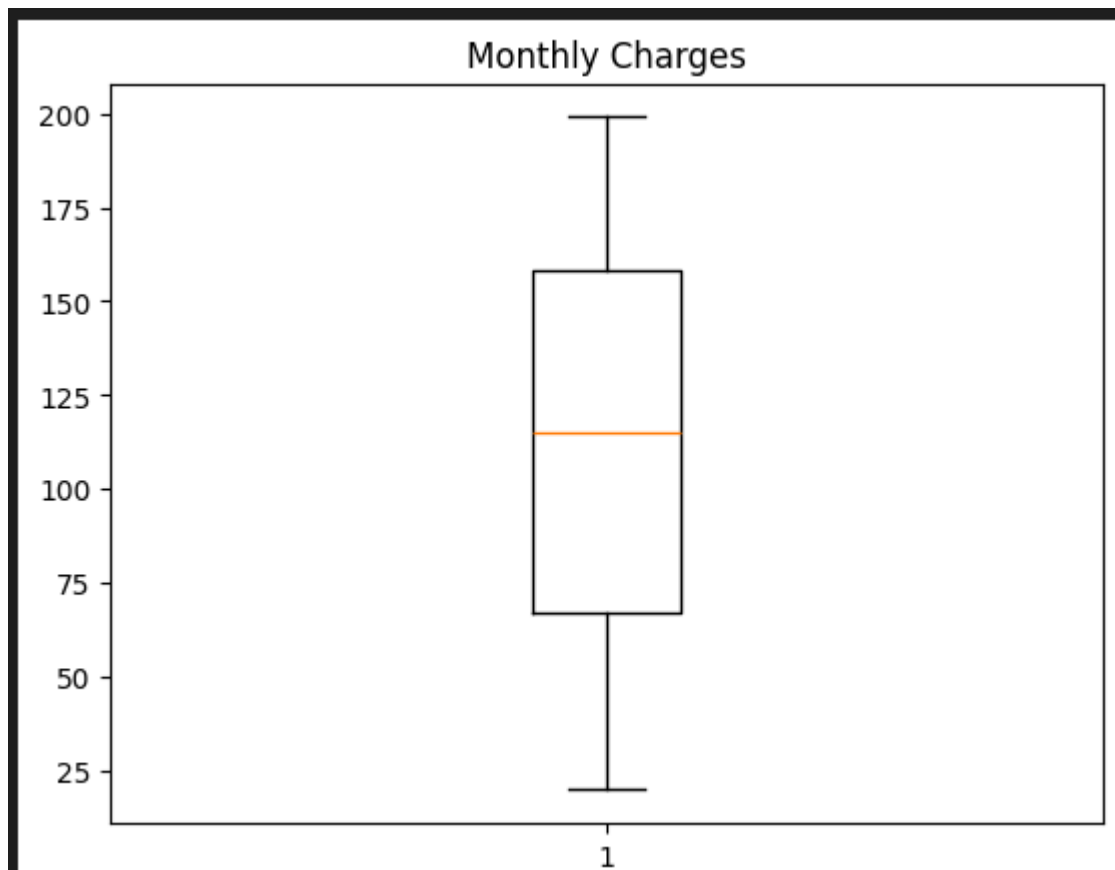


- Pie Chart

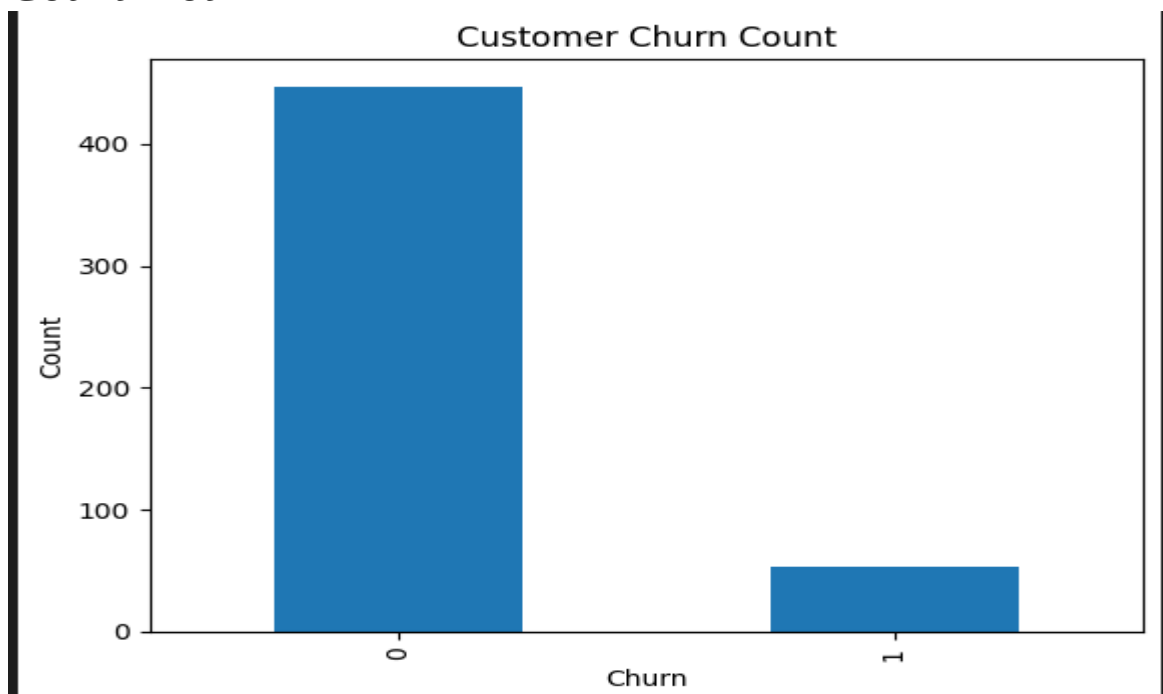


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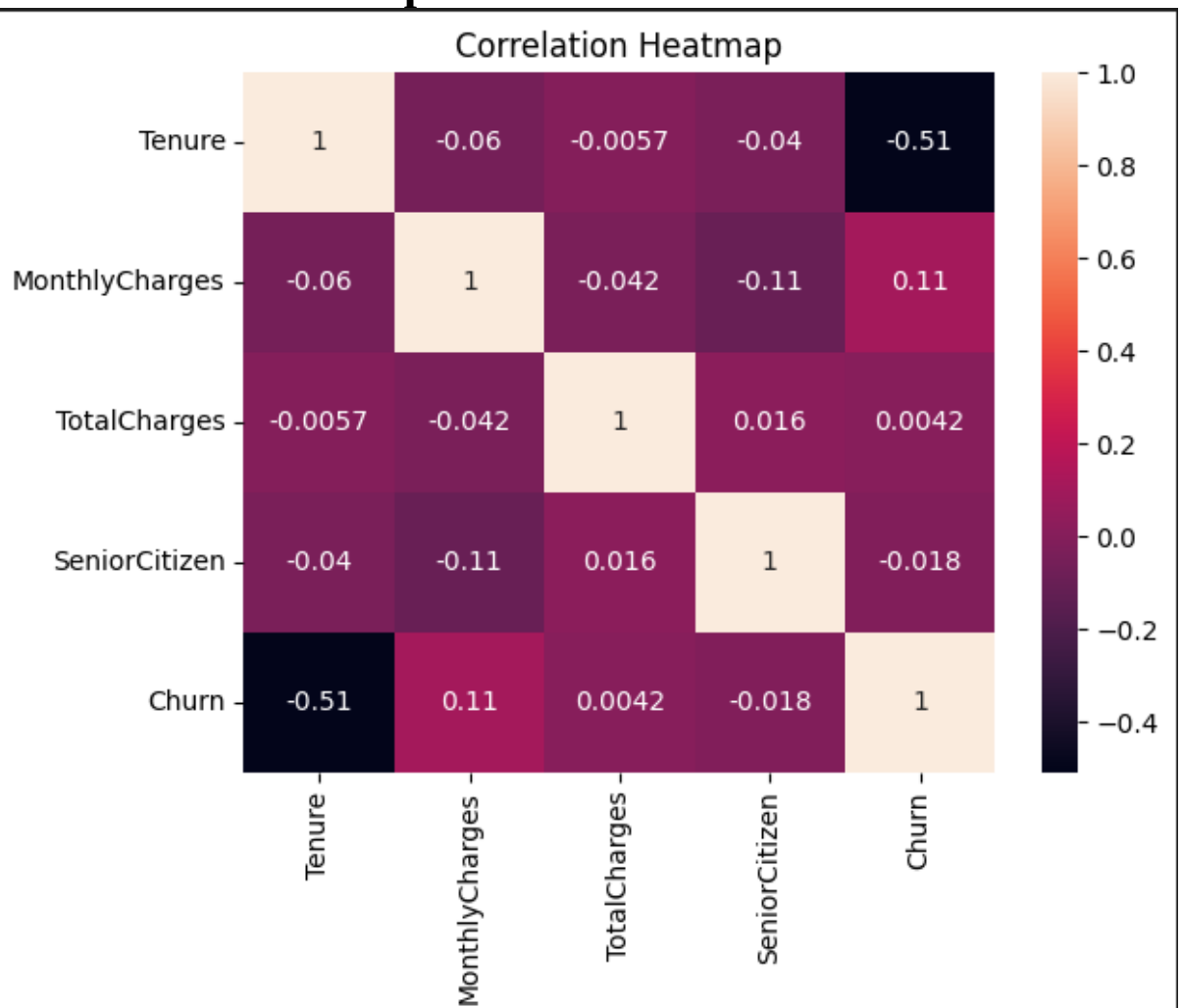
- **Box Plot**



- **Count Plot**



- **Correlation Heatmap**



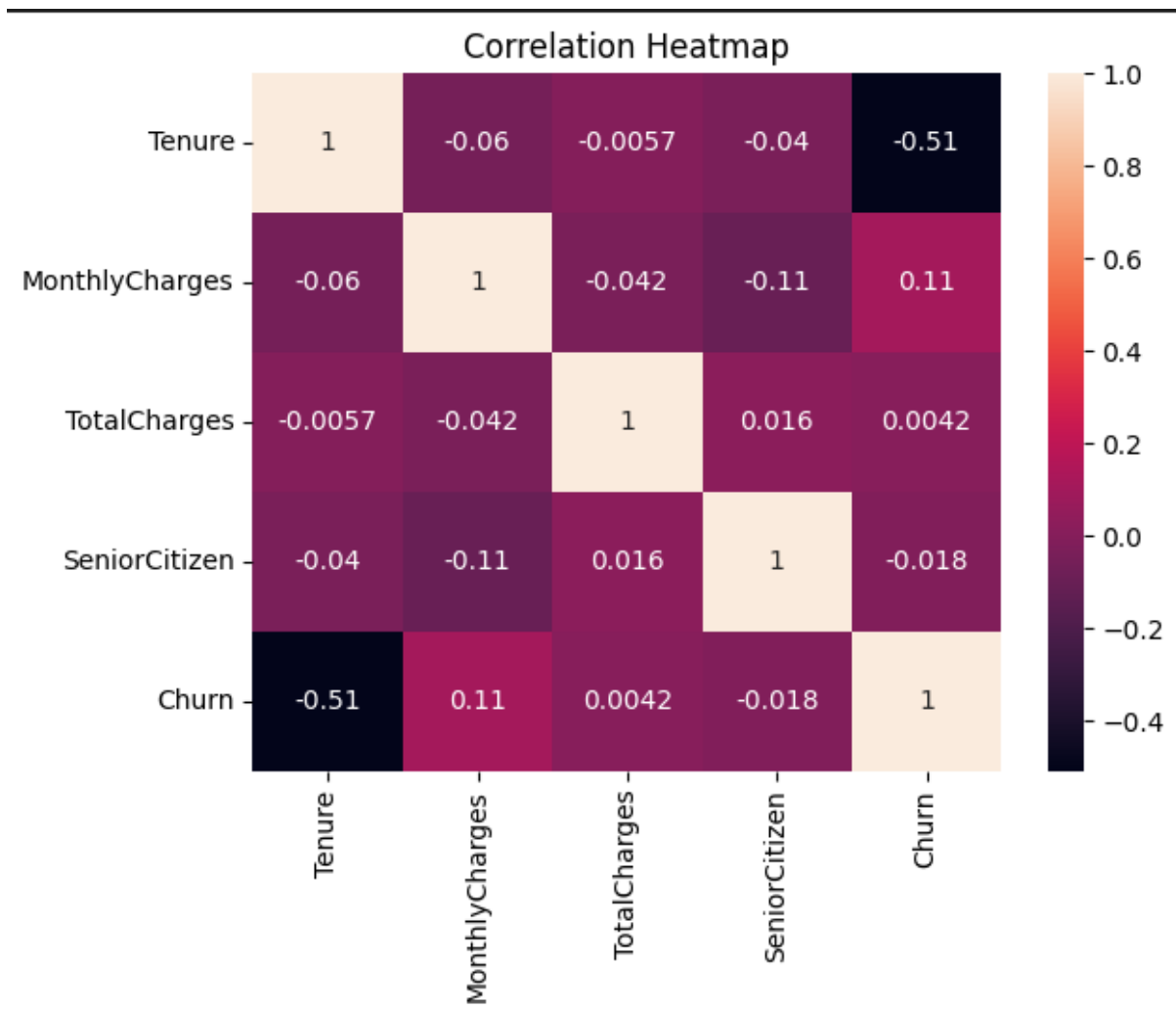
These visualizations provide a graphical representation of customer churn and business patterns.

10. Correlation Analysis

Correlation analysis was performed to understand the relationship between numerical variables.

A correlation heatmap was generated to visualize positive and negative relationships among variables.

This helps identify which variables influence each other.



11. Hypothesis Testing

A T-Test was performed to compare Monthly Charges between churned and non-churned customers.

Hypotheses:

H_0 (Null Hypothesis):

There is no significant difference in Monthly Charges between churned and retained customers.

H_1 (Alternative Hypothesis):

There is a significant difference in Monthly Charges between churned and retained customers.

The p-value was used to determine statistical significance

12. Regression Analysis

A Linear Regression model was built.

Independent Variable

- Tenure

Dependent Variable

- Monthly Charges

The regression model was used to understand the relationship between customer tenure and monthly charges.

13. Business Insights

The analysis generated several important business insights.

- Customer churn affects company revenue.
- Higher monthly charges may increase customer churn.
- Long-term customers are generally more loyal.
- Customer satisfaction plays an important role in customer retention.
- Data analysis helps identify customers at higher risk of leaving.

These insights support better business decision-making.

14. Recommendations

Based on the analysis, the following recommendations are suggested:

- Improve customer support services.
- Introduce customer loyalty programs.
- Offer personalized discounts.
- Reduce monthly charges for selected customer groups.
- Conduct regular customer satisfaction surveys.
- Monitor churn rates monthly.
- Improve communication with customers.
- Reward long-term customers.

Implementing these recommendations can improve customer retention and business performance.

15. Conclusion

This project successfully demonstrated the complete data analysis workflow using Python.

The customer churn dataset was cleaned, analyzed, and visualized. Statistical methods including correlation analysis, hypothesis testing, and regression analysis were performed successfully.

The project generated meaningful business insights and practical recommendations that can help businesses reduce customer churn and improve customer satisfaction.

Overall, this project demonstrates how data-driven decision-making supports business growth and operational efficiency.

16. Future Scope

Future improvements include:

- Machine Learning Prediction Models
- Customer Segmentation
- Interactive Power BI Dashboard
- Real-Time Data Analysis
- Customer Lifetime Value Prediction
- Churn Prediction System
- Web Dashboard Development

These enhancements can make the analysis more accurate and useful for businesses.

17. References

1. Python Official Documentation
2. Pandas Documentation
3. NumPy Documentation
4. Matplotlib Documentation
5. Seaborn Documentation
6. Scikit-Learn Documentation
7. Customer Churn Dataset

